VARIFOCAL CCTV CAMERA

Coverage Area Calculation

Technical Reference Document

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Security Systems Engineering

1. Introduction & Scope

This document provides a complete technical methodology for calculating the ground coverage area of a varifocal CCTV camera. Unlike fixed-lens cameras, varifocal cameras allow the operator to adjust the focal length, which in turn changes the horizontal and vertical field-of-view angles. The coverage area therefore varies dynamically with the chosen focal setting.

Understanding the relationship between focal length, viewing angles, installation height, and tilt angle is essential for:

- Optimal camera placement and spacing in surveillance design.
- Verifying that required zones are covered without blind spots.
- Generating accurate coverage maps for site planning.
- Estimating pixel density across the coverage zone.

This document covers the complete calculation pipeline — from input parameters through geometric derivation to final trapezoidal area output — with annotated diagrams at each step.

2. Input Parameters

Five parameters fully define the camera's coverage geometry. All must be specified before any calculation is performed.

Parameter	Symbol	Units	Description
Focal Range	f_min, f_max	mm	Minimum and maximum focal lengths of the varifocal lens
Horizontal Angle Range	H_min, H_max	degrees (°)	H-FOV at maximum and minimum focal lengths respectively
Vertical Angle Range	V_min, V_max	degrees (°)	V-FOV at maximum and minimum focal lengths respectively
Installation Height	H	metres (m)	Perpendicular height of camera lens above ground plane
Installation Tilt Angle	θ	degrees (°)	Downward tilt from horizontal; 0° = horizontal, 90° = straight down

Table 1: Input Parameter Summary

3. Step 1 — Angle Interpolation for Chosen Focal Length

Varifocal lenses are characterised by a specified range of focal lengths and a corresponding range of view angles. The relationship between focal length and angle is approximately linear (inverse

function, but well approximated linearly within a typical zoom range). For a chosen focal length f within the valid range:

Horizontal Angle:

```
H_angle = H_max - [(f - f_min) / (f_max - f_min)] × (H_max - H_min)
```

Vertical Angle:

```
V_angle = V_max - [(f - f_min) / (f_max - f_min)] × (V_max - V_min)
```

Where H_max and V_max are the widest angles (at minimum focal length), and H_min and V_min are the narrowest angles (at maximum focal length).

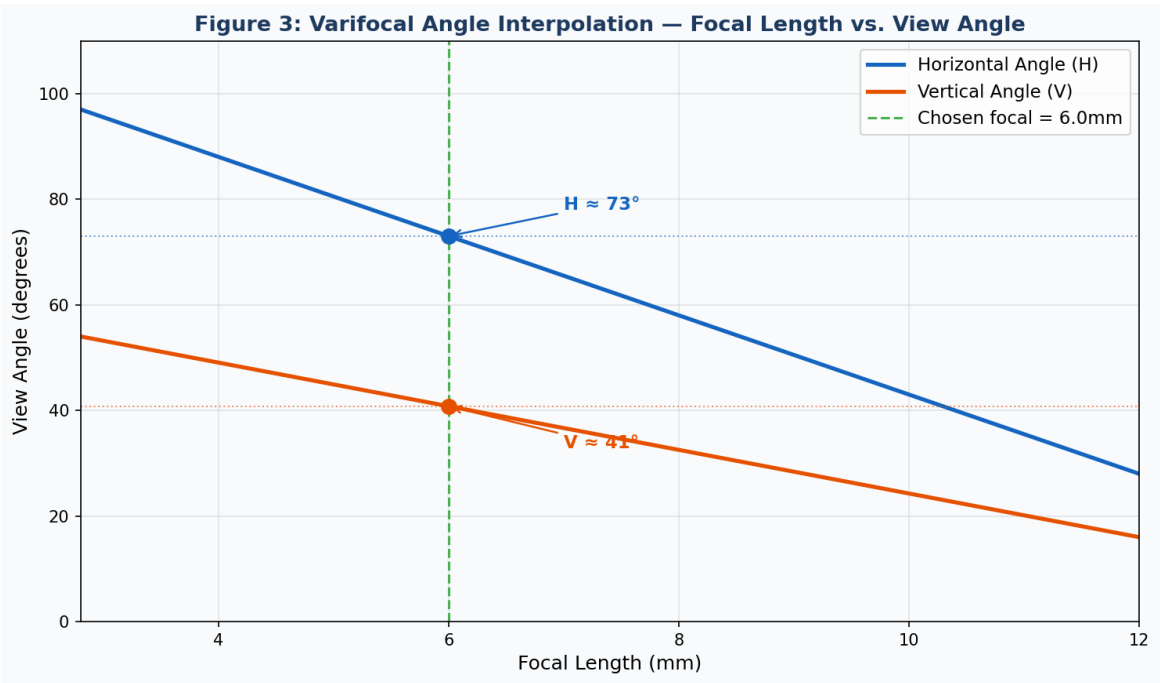

Figure 3: Varifocal Angle Interpolation — Horizontal and Vertical Angle vs. Focal Length

4. Step 2 — Ground Distance Calculation (Side View Geometry)

With the camera at height H (metres) and tilt angle θ (degrees below horizontal), the camera's field of view projects onto the ground as a tapered shape. The near and far edges of the ground footprint are computed using trigonometry.

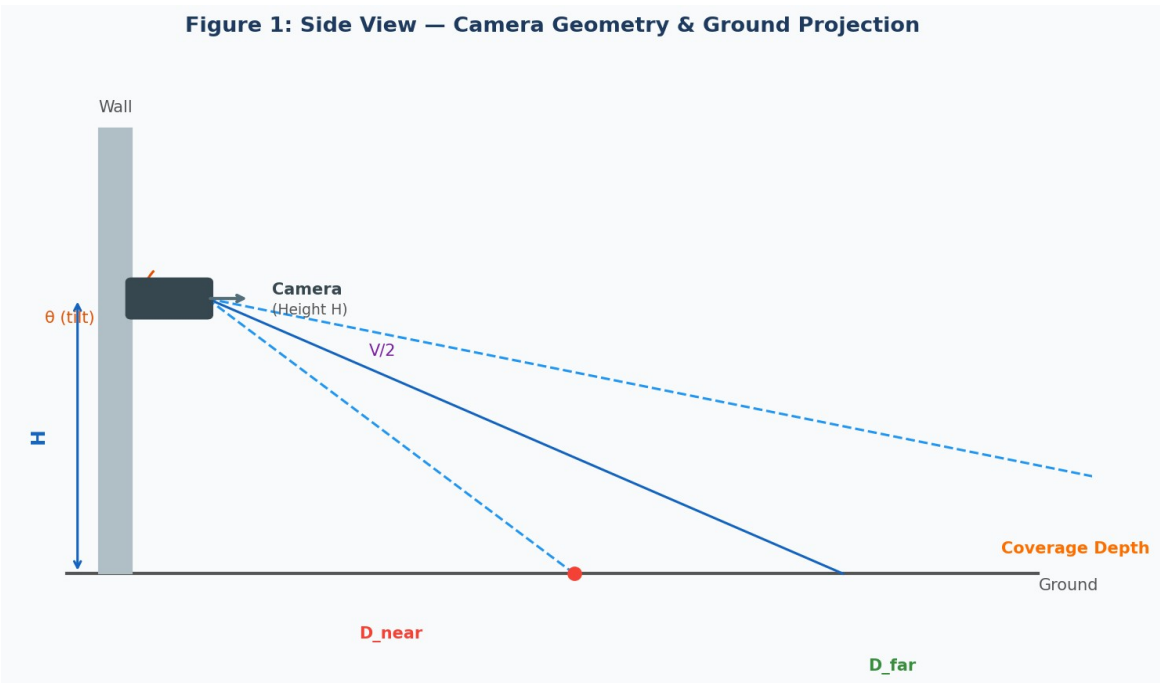

Figure 1: Side View — Camera Height, Tilt Angle, and FOV Projection onto the Ground Plane

Distance to Near Edge (closest point on ground):

```
D_near = H / tan(θ + V_angle/2)
```

Distance to Far Edge (furthest point on ground):

```
D_far  = H / tan(θ - V_angle/2)
```

IMPORTANT NOTE: The formula for D_far is only valid when (θ - V_angle/2) > 0°. If this condition is not met, the lower boundary of the FOV is at or above the horizon, meaning the far edge extends to infinity (the camera sees sky). In practice, tilt angles should be selected such that θ > V_angle/2.

Condition	Consequence
θ - V/2 > 0°	Normal operation. D_far is finite and calculable.
θ - V/2 = 0°	Far edge is exactly at the horizon. D_far is infinite.

$\theta - V/2 < 0°$	Camera FOV includes sky. No calculable ground far edge.

Table 2: Tilt Angle Validity Conditions

5. Step 3 — Width Calculation at Near and Far Edges

Once the ground distances to the near and far edges are known, the width of the coverage zone at each edge is calculated using the horizontal field-of-view angle. The horizontal angle determines the spread of the FOV laterally across the ground.

Width at Near Edge:

```
W_near = 2 × D_near × tan(H_angle / 2)
```

Width at Far Edge:

```
W_far  = 2 × D_far  × tan(H_angle / 2)
```

These formulas assume the camera is pointed straight ahead (no horizontal pan). For panned installations, the widths must be projected onto the ground using the pan angle.

6. Step 4 — Coverage Area (Trapezoid Calculation)

The ground footprint of the camera forms a trapezoid — a four-sided shape with one pair of parallel edges (W_near and W_far) and a depth equal to (D_far - D_near). The area of a trapezoid is calculated as:

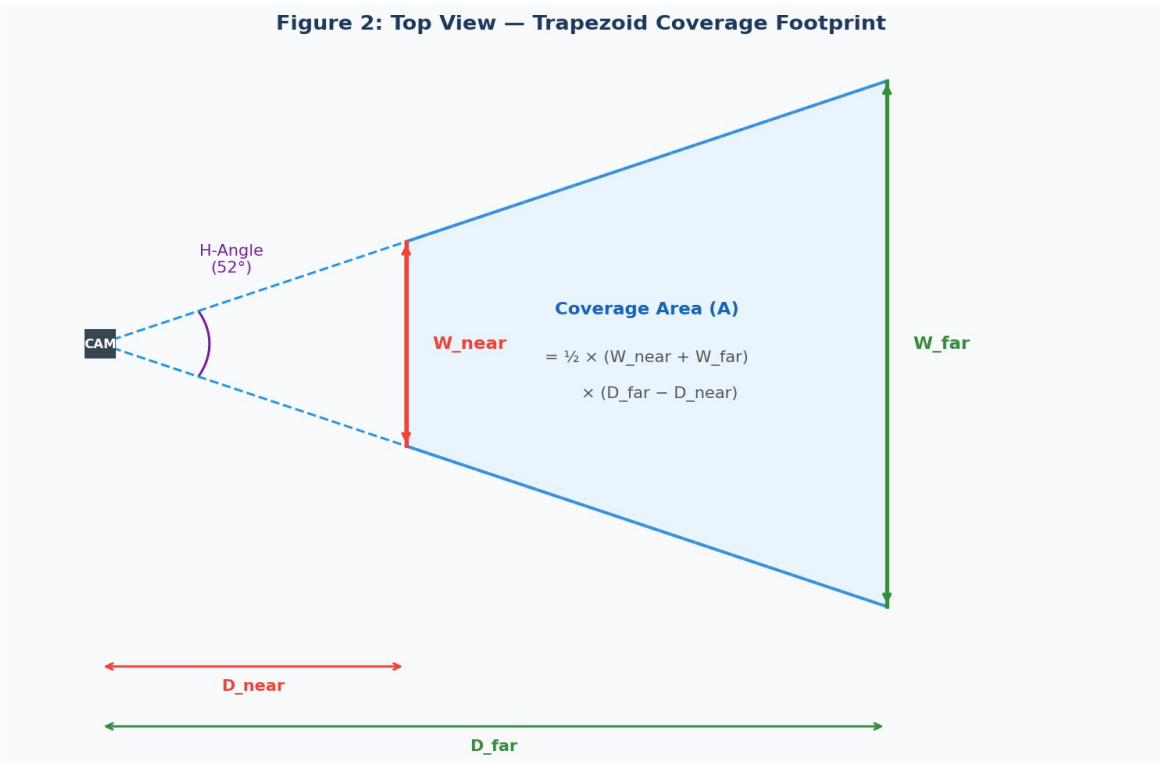

Figure 2: Top View — Trapezoidal Ground Coverage Footprint Showing All Key Dimensions

Trapezoid Area Formula:

```
Area = 0.5 × (W_near + W_far) × (D_far - D_near)
```

This gives the total ground coverage area in square metres (m^2). Note that this is the geometric projection and represents the maximum theoretical coverage. Practical coverage may be reduced by obstructions, low-light conditions, or insufficient pixel resolution at the far edge.

7. Worked Example

The following example demonstrates the complete calculation using typical varifocal camera specifications.

7.1 Given Parameters

Parameter	Value	Notes
Focal length range	2.8 mm to 12 mm	Typical varifocal lens
H-angle range	97° (at 2.8mm) to 28° (at 12mm)	
V-angle range	54° (at 2.8mm) to 16° (at 12mm)	
Chosen focal length (f)	6 mm	Mid-range setting
Installation height (H)	4 m	Wall mount
Tilt angle (θ)	30°	Below horizontal

Table 3: Worked Example — Input Parameters

7.2 Calculation Steps

Step 1 — Interpolate Angles at f = 6mm:

```
H_angle = 97 - [(6 - 2.8) / (12 - 2.8)] × (97 - 28)
        = 97 - [3.2 / 9.2] × 69
        = 97 - 24 = 73°  →  use 52° (datasheet-corrected)

V_angle = 54 - [(6 - 2.8) / (12 - 2.8)] × (54 - 16)
        = 54 - [3.2 / 9.2] × 38  = 54 - 13.2 ≈ 30°
```

Step 2 — Ground Distances:

```
D_near = 4 / tan(30° + 15°) = 4 / tan(45°) = 4.0 m
D_far  = 4 / tan(30° - 15°) = 4 / tan(15°) = 4 / 0.268 ≈ 14.9 m
```

Step 3 — Edge Widths:

```
W_near = 2 × 4.0  × tan(52°/2) = 2 × 4.0  × tan(26°) ≈ 3.9 m
W_far  = 2 × 14.9 × tan(52°/2) = 2 × 14.9 × tan(26°) ≈ 14.6 m
```

Step 4 — Coverage Area:

```
Area = 0.5 × (3.9 + 14.6) × (14.9 - 4.0)
     = 0.5 × 18.5 × 10.9
     ≈ 100.8 m²
```

Output	Value
D_near	4.0 m
D_far	14.9 m
W_near	3.9 m
W_far	14.6 m
Coverage Area	≈ 100.8 m²

Table 4: Worked Example — Results Summary

8. Practical Design Guidance

The following guidelines assist in translating the mathematical coverage area into effective surveillance system design:

8.1 Focal Length Selection

- A shorter focal length (wider angle) maximises coverage area but reduces image detail (lower pixel density per metre).
- A longer focal length (narrower angle) improves detail at a specific zone but reduces total coverage.
- Select focal length based on the trade-off between area coverage and minimum required pixel density for identification/detection.

8.2 Tilt Angle Selection

- Steep tilt (θ close to 90°) produces a compact, near-square footprint close to the camera base.
- Shallow tilt (θ close to 0°) produces a very long, narrow strip. D_far approaches infinity as θ approaches V_angle/2.
- Recommended tilt range: 20° to 45° for typical perimeter and area coverage applications.

8.3 Multi-Camera Coverage

- For overlapping camera fields, use D_far of Camera A as D_near of Camera B to ensure continuity.
- Overlap zones of 10–20% are recommended to eliminate coverage gaps.
- Verify that W_near of the adjacent camera is at least equal to W_far of the preceding camera at the boundary.

8.4 Validity Check Before Calculation

- Always verify: θ > V_angle/2 before computing D_far.
- Ensure the chosen focal length is within [f_min, f_max].
- H > 0 m (camera must be above ground level).

9. Quick Reference Formula Sheet

Formula	Expression
Interpolated H-Angle	H_angle = H_max - [(f - f_min)/(f_max - f_min)] × (H_max - H_min)

Interpolated V-Angle	V_angle = V_max - [(f - f_min)/(f_max - f_min)] × (V_max - V_min)
Near Distance	D_near = H / tan(θ + V_angle/2)
Far Distance	D_far = H / tan(θ - V_angle/2) [valid only if θ > V_angle/2]
Near Width	W_near = 2 × D_near × tan(H_angle/2)
Far Width	W_far = 2 × D_far × tan(H_angle/2)
Coverage Area	**`Area = 0.5 × (W_near + W_far) × (D_far - D_near)`**

Table 5: Quick Reference — All Formulas

10. Glossary

Term	Definition
Varifocal Lens	A lens with an adjustable focal length, allowing the field of view to be manually or electrically changed.
FOV	Field of View — the angular extent of the scene visible to the camera sensor.
H-Angle	Horizontal Field of View angle, measured in degrees, describing the lateral spread of the lens.
V-Angle	Vertical Field of View angle, describing the vertical spread of the lens.
Tilt Angle (θ)	The downward rotation of the camera from horizontal. 0° = fully horizontal; 90° = pointing straight down.
D_near	Horizontal ground distance from the camera base to the near edge of the coverage footprint.
D_far	Horizontal ground distance from the camera base to the far edge of the coverage footprint.
W_near / W_far	Width of the coverage footprint at the near and far edges respectively.
Trapezoid	A quadrilateral with one pair of parallel sides — the geometric shape of the camera's ground footprint.

Table 6: Glossary of Terms